

# KIT SAFETY DATA SHEET



Kit Product Name Bio-Plex Pro Human Chemokine Panel

Kit Catalogue Number(s) 171AK99MR2, MZ0000XVVLFGWM

Revision date 13-Feb-2024

## Kit Contents

| Catalogue Number(s)                                                 | Product Name                                                   |
|---------------------------------------------------------------------|----------------------------------------------------------------|
| 9723892, 9703892, 9704415, 10014822, 10014823                       | Bio-Plex Assay Buffer                                          |
| 171304040, 10027955, 12006121, 12005850                             | Bio-Plex Pro Assays 10X Wash Buffer                            |
| 171DK0001, 171DK0050, 10032534                                      | Bio-Plex Pro Human Chemokine Standards                         |
| 10032535                                                            | Bio-Plex Pro Human Chemokine Panel 40-plex Lyophilized Control |
| 10032533                                                            | Bio-Plex Pro Human Chemokine Detection Antibodies              |
| 10032444                                                            | Bio-Plex Pro Human Chemokine 40-Plex Beads                     |
| 10032400, 10031831, 12005852                                        | Bio-Plex Detection Antibody Diluent HB                         |
| 10014641, 9704423, 9703895, 171305043, 10022628, 10041561, 12005851 | Bio-Plex Sample Diluent                                        |
| 9703888, 9704424, 171305042, 171304080M, 10022368, 12005853         | Bio-Plex Standard Diluent                                      |
| 171304501, 9704418, 9703887, 9703897                                | Streptavidin-PE                                                |

# SAFETY DATA SHEET

Revision date 14-Nov-2023

Revision Number 1.2

## Section 1: Identification

### Product identifier

**Product Name** Bio-Plex Assay Buffer  
**Catalogue Number(s)** 9723892, 9703892, 9704415, 10014822, 10014823

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of the supplier of the safety data sheet

|                                                                                                      |                                                                                                                                 |                                                                                             |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Supplier</b><br>Bio-Rad Laboratories Inc.<br>1000 Alfred Nobel Drive<br>Hercules, CA 94547<br>USA | <b>Manufacturer</b><br>Bio-Rad Laboratories, Life Science Group<br>2000 Alfred Nobel Drive<br>Hercules, California 94547<br>USA | <b>Importer</b><br>Bio-Rad Laboratories Pty Ltd<br>189 Bush Road<br>Auckland<br>New Zealand |
| <b>Technical Service</b>                                                                             | +64 9 415 2280 or 0508 805 500<br>sales.nz@bio-rad.com                                                                          |                                                                                             |

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS) Not classified

### Label elements

#### **Hazard statements**

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS) Not classified

### Other hazards which do not result in classification

No information available.

## Section 3: Composition/information on ingredients

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Non-hazardous ingredients | Proprietary | Balance  |

## Section 4: First-aid measures

### Description of first aid measures

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                              |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| Skin contact | Wash skin with soap and water.                                                                                    |
| Ingestion    | Rinse mouth.                                                                                                      |

### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

### Indication of any immediate medical attention and special treatment needed

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Large Fire                     | CAUTION: Use of water spray when fighting fire may be inefficient.                                      |
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams.                                       |

### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

### Special protective actions for fire-fighters

|                                                                |                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|

## Section 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                          |                                                   |
|--------------------------|---------------------------------------------------|
| Personal precautions     | Ensure adequate ventilation.                      |
| For emergency responders | Use personal protection recommended in Section 8. |

### Environmental precautions

|                           |                                                       |
|---------------------------|-------------------------------------------------------|
| Environmental precautions | See Section 12 for additional Ecological Information. |
|---------------------------|-------------------------------------------------------|

### Methods and material for containment and cleaning up

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
|-------------------------|-------------------------------------------------------|

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**Incompatible materials** Metals.

## Section 8: Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** No special protective equipment required.

**Hand protection** No special protective equipment required.

**Skin and body protection** No special protective equipment required.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## Section 9: Physical and chemical properties

**Information on basic physical and chemical properties**

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                         | <u>Values</u>             | <u>Remarks • Method</u> |
|-----------------------------------------|---------------------------|-------------------------|
| pH                                      | 7.4                       |                         |
| Melting point / freezing point          | 0 °C                      |                         |
| Initial boiling point and boiling range | 100 °C                    |                         |
| Flash point                             | No data available         | None known              |
| Evaporation rate                        | No data available         | None known              |
| Flammability                            | No data available         | None known              |
| Flammability Limit in Air               |                           | None known              |
| Upper flammability or explosive limits  | No data available         |                         |
| Lower flammability or explosive limits  | No data available         |                         |
| Vapour pressure                         | No data available         | None known              |
| Relative vapour density                 | No data available         | None known              |
| Relative density                        | No data available         | None known              |
| Water solubility                        | Miscible in water         |                         |
| Solubility(ies)                         | No data available         | None known              |
| Partition coefficient                   | No data available         | None known              |
| Autoignition temperature                | No data available         | None known              |
| Decomposition temperature               |                           | None known              |
| Kinematic viscosity                     | No data available         | None known              |
| Dynamic viscosity                       | No data available         | None known              |
| Explosive properties                    | No information available. |                         |
| Oxidising properties                    | No information available. |                         |
| <u>Other information</u>                |                           |                         |
| Softening point                         | No information available  |                         |
| Molecular weight                        | No information available  |                         |
| VOC content                             | No information available  |                         |
| Liquid Density                          | No information available  |                         |
| Bulk density                            | No information available  |                         |
| Particle characteristics                | No information available  |                         |

## Section 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions Avoid contact with metals. This product contains Sodium azide. Sodium azide can react with Copper, Brass, Lead, and solder in piping systems to form explosive compounds and toxic gases.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials Metals.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**Section 11: Toxicological information****Acute toxicity****Information on likely routes of exposure****Product Information**

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms** No information available.

**Acute toxicity**

.

**Numerical measures of toxicity****Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitisation** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Data used to identify the health effects** Refer to Section 16 for Key literature references and sources for data used to compile the SDS.

**Section 12: Ecological information**

**Ecotoxicity**

|                                      |                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Aquatic ecotoxicity</b>           | The environmental impact of this product has not been fully investigated.                  |
| <b>Unknown aquatic toxicity</b>      | 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment. |
| <b>Terrestrial ecotoxicity</b>       | There is no data for this product.                                                         |
| <b>Persistence and degradability</b> | No information available.                                                                  |

**Bioaccumulative potential**

|                        |                                    |
|------------------------|------------------------------------|
| <b>Bioaccumulation</b> | There is no data for this product. |
|------------------------|------------------------------------|

**Mobility in soil**

|                 |                           |
|-----------------|---------------------------|
| <b>Mobility</b> | No information available. |
|-----------------|---------------------------|

**Other adverse effects**

No information available.

**Section 13: Disposal considerations****Disposal methods**

|                                            |                                                                                                                                                                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Not applicable.<br>Not Hazardous.<br>Dispose of in accordance with local regulations.<br>Dispose of waste in accordance with environmental legislation.<br>Flush pipes with water frequently if discarding solutions containing Sodium azide into metal piping systems. |
| <b>Contaminated packaging</b>              | Not applicable.<br>Not Hazardous.                                                                                                                                                                                                                                       |

**Section 14: Transport information**

|                    |               |
|--------------------|---------------|
| <b><u>IATA</u></b> | Not regulated |
|--------------------|---------------|

|                    |               |
|--------------------|---------------|
| <b><u>IMDG</u></b> | Not regulated |
|--------------------|---------------|

**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available

**Special precautions for user**  
Please refer to the applicable dangerous goods regulations for additional information

**Section 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations**

**EPA New Zealand HSNO approval** To be determined

**code or group standard****National regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

**NZIoC**

Contact supplier for inventory compliance status.

**TSCA**

Contact supplier for inventory compliance status.

**DSL/NDSL**

Contact supplier for inventory compliance status.

**EINECS/ELINCS**

Contact supplier for inventory compliance status.

**ENCS**

Contact supplier for inventory compliance status.

**IECSC**

Contact supplier for inventory compliance status.

**KECL**

Contact supplier for inventory compliance status.

**PICCS**

Contact supplier for inventory compliance status.

**AIIC**

Contact supplier for inventory compliance status.

**Legend:**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

**ENCS** - Japan Existing and New Chemical Substances

**IECSC** - China Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**Section 16: Other information****Revision date**

14-Nov-2023

**Revision Note**

Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)

U.S. Environmental Protection Agency ChemView Database

European Food Safety Authority (EFSA)

EPA (Environmental Protection Agency)

Acute Exposure Guideline Level(s) (AEGl(s))

U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act



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U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

**End of Safety Data Sheet**

# SAFETY DATA SHEET

Revision date 07-Feb-2024

Revision Number 2

## Section 1: Identification

### Product identifier

**Product Name** Bio-Plex Pro Assays 10X Wash Buffer  
**Catalogue Number(s)** 171304040, 10027955, 12006121, 12005850

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of the supplier of the safety data sheet

|                                                                                                      |                                                                                                                                 |                                                                                             |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Supplier</b><br>Bio-Rad Laboratories Inc.<br>1000 Alfred Nobel Drive<br>Hercules, CA 94547<br>USA | <b>Manufacturer</b><br>Bio-Rad Laboratories, Life Science Group<br>2000 Alfred Nobel Drive<br>Hercules, California 94547<br>USA | <b>Importer</b><br>Bio-Rad Laboratories Pty Ltd<br>189 Bush Road<br>Auckland<br>New Zealand |
| <b>Technical Service</b>                                                                             | +64 9 415 2280 or 0508 805 500<br>sales.nz@bio-rad.com                                                                          |                                                                                             |

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS) Not classified

### Label elements

#### **Hazard statements**

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS) Not classified

### Other hazards which do not result in classification

No information available.

## Section 3: Composition/information on ingredients

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Non-hazardous ingredients | Proprietary | Balance  |

## Section 4: First-aid measures

### Description of first aid measures

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                              |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| Skin contact | Wash skin with soap and water.                                                                                    |
| Ingestion    | Rinse mouth.                                                                                                      |

### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

### Indication of any immediate medical attention and special treatment needed

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Large Fire                     | CAUTION: Use of water spray when fighting fire may be inefficient.                                      |
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams.                                       |

### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

### Special protective actions for fire-fighters

|                                                                |                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|

## Section 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                          |                                                   |
|--------------------------|---------------------------------------------------|
| Personal precautions     | Ensure adequate ventilation.                      |
| For emergency responders | Use personal protection recommended in Section 8. |

### Environmental precautions

|                           |                                                       |
|---------------------------|-------------------------------------------------------|
| Environmental precautions | See Section 12 for additional Ecological Information. |
|---------------------------|-------------------------------------------------------|

### Methods and material for containment and cleaning up

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
|-------------------------|-------------------------------------------------------|

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store according to product and label instructions.

**Incompatible materials** Metals.

## Section 8: Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** No special protective equipment required.

**Hand protection** No special protective equipment required.

**Skin and body protection** No special protective equipment required.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## Section 9: Physical and chemical properties

**Information on basic physical and chemical properties**

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                         | <u>Values</u>             | <u>Remarks • Method</u> |
|-----------------------------------------|---------------------------|-------------------------|
| pH                                      | 7.4                       |                         |
| Melting point / freezing point          | No data available         | None known              |
| Initial boiling point and boiling range | 100 °C                    |                         |
| Flash point                             | No data available         | None known              |
| Evaporation rate                        | No data available         | None known              |
| Flammability                            | No data available         | None known              |
| Flammability Limit in Air               |                           | None known              |
| Upper flammability or explosive limits  | No data available         |                         |
| Lower flammability or explosive limits  | No data available         |                         |
| Vapour pressure                         | No data available         | None known              |
| Relative vapour density                 | No data available         | None known              |
| Relative density                        | No data available         | None known              |
| Water solubility                        | Miscible in water         |                         |
| Solubility(ies)                         | No data available         | None known              |
| Partition coefficient                   | No data available         | None known              |
| Autoignition temperature                | No data available         | None known              |
| Decomposition temperature               |                           | None known              |
| Kinematic viscosity                     | No data available         | None known              |
| Dynamic viscosity                       | No data available         | None known              |
| Explosive properties                    | No information available. |                         |
| Oxidising properties                    | No information available. |                         |
| <u>Other information</u>                |                           |                         |
| Softening point                         | No information available  |                         |
| Molecular weight                        | No information available  |                         |
| VOC content                             | No information available  |                         |
| Liquid Density                          | No information available  |                         |
| Bulk density                            | No information available  |                         |
| Particle characteristics                | No information available  |                         |

## Section 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions Avoid contact with metals. This product contains Sodium azide. Sodium azide can react with Copper, Brass, Lead, and solder in piping systems to form explosive compounds and toxic gases.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials Metals.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

**Section 11: Toxicological information****Acute toxicity****Information on likely routes of exposure****Product Information**

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Inhalation   | Specific test data for the substance or mixture is not available. |
| Eye contact  | Specific test data for the substance or mixture is not available. |
| Skin contact | Specific test data for the substance or mixture is not available. |
| Ingestion    | Specific test data for the substance or mixture is not available. |

**Symptoms** No information available.

**Acute toxicity**

.

**Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 34,207.50 mg/kg

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitisation** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Data used to identify the health effects** Refer to Section 16 for Key literature references and sources for data used to compile the SDS.

**Section 12: Ecological information**

**Ecotoxicity**

|                                      |                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Aquatic ecotoxicity</b>           | The environmental impact of this product has not been fully investigated.                  |
| <b>Unknown aquatic toxicity</b>      | 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment. |
| <b>Terrestrial ecotoxicity</b>       | There is no data for this product.                                                         |
| <b>Persistence and degradability</b> | No information available.                                                                  |

**Bioaccumulative potential**

|                        |                                    |
|------------------------|------------------------------------|
| <b>Bioaccumulation</b> | There is no data for this product. |
|------------------------|------------------------------------|

**Mobility in soil**

|                 |                           |
|-----------------|---------------------------|
| <b>Mobility</b> | No information available. |
|-----------------|---------------------------|

**Other adverse effects**

No information available.

**Section 13: Disposal considerations****Disposal methods**

|                                            |                                                                                                                                                                                                                                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from residues/unused products</b> | Not applicable.<br>Not Hazardous.<br>Dispose of in accordance with local regulations.<br>Dispose of waste in accordance with environmental legislation.<br>Flush pipes with water frequently if discarding solutions containing Sodium azide into metal piping systems. |
| <b>Contaminated packaging</b>              | Not applicable.<br>Not Hazardous.                                                                                                                                                                                                                                       |

**Section 14: Transport information**

|                    |               |
|--------------------|---------------|
| <b><u>IATA</u></b> | Not regulated |
|--------------------|---------------|

|                    |               |
|--------------------|---------------|
| <b><u>IMDG</u></b> | Not regulated |
|--------------------|---------------|

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information

**Section 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations**

**EPA New Zealand HSNO approval code or group standard** To be determined

**National regulations** There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements** Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information  
Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information  
Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

### International Inventories

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDSL</b>      | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AIIC</b>          | Contact supplier for inventory compliance status. |

#### **Legend:**

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances  
**ENCS** - Japan Existing and New Chemical Substances  
**IECSC** - China Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances

## Section 16: Other information

**Revision date** 07-Feb-2024

**Revision Note** SDS sections updated. 2. 12.

### **Key or legend to abbreviations and acronyms used in the safety data sheet**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|                |                             |             |                                  |
|----------------|-----------------------------|-------------|----------------------------------|
| <b>TWA</b>     | TWA (time-weighted average) | <b>STEL</b> | STEL (Short Term Exposure Limit) |
| <b>Ceiling</b> | Maximum limit value         | <b>*</b>    | Skin designation                 |
| <b>C</b>       | Carcinogen                  |             |                                  |

### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGL(s))



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U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

**End of Safety Data Sheet**

# SAFETY DATA SHEET

Revision date 13-Feb-2024

Revision Number 1.2

## Section 1: Identification

### Product identifier

**Product Name** Bio-Plex Pro Human Chemokine Standards

**Catalogue Number(s)** 171DK0001, 171DK0050, 10032534

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Supplier

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Importer

Bio-Rad Laboratories Pty Ltd  
189 Bush Road  
Auckland  
New Zealand

**Technical Service** +64 9 415 2280 or 0508 805 500  
sales.nz@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Label elements

#### **Hazard statements**

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Other hazards which do not result in classification

No information available.

## Section 3: Composition/information on ingredients

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Non-hazardous ingredients | Proprietary | Balance  |

## Section 4: First-aid measures

### Description of first aid measures

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| Skin contact | Wash skin with soap and water.                                                                                       |
| Ingestion    | Rinse mouth.                                                                                                         |

### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

### Indication of any immediate medical attention and special treatment needed

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media   | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Large Fire                     | CAUTION: Use of water spray when fighting fire may be inefficient.                                      |
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams.                                       |

### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

### Special protective actions for fire-fighters

|                                                                |                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|

## Section 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                          |                                                   |
|--------------------------|---------------------------------------------------|
| Personal precautions     | Ensure adequate ventilation.                      |
| For emergency responders | Use personal protection recommended in Section 8. |

### Environmental precautions

|                           |                                                       |
|---------------------------|-------------------------------------------------------|
| Environmental precautions | See Section 12 for additional Ecological Information. |
|---------------------------|-------------------------------------------------------|

### Methods and material for containment and cleaning up

|                         |                                                       |
|-------------------------|-------------------------------------------------------|
| Methods for containment | Prevent further leakage or spillage if safe to do so. |
|-------------------------|-------------------------------------------------------|

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** For long-term storage, keep the battery at temperatures between 10°C-30°C.

**Incompatible materials** None known based on information supplied.

## Section 8: Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** No special protective equipment required.

**Hand protection** No special protective equipment required.

**Skin and body protection** No special protective equipment required.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## Section 9: Physical and chemical properties

**Information on basic physical and chemical properties**

**Physical state** Solid  
**Appearance** powder or cake, lyophilised  
**Colour** white  
**Odour** Odourless.  
**Odour threshold** No information available

| <u>Property</u>                         | <u>Values</u>             | <u>Remarks • Method</u> |
|-----------------------------------------|---------------------------|-------------------------|
| pH                                      | 7.4                       |                         |
| Melting point / freezing point          | No data available         | None known              |
| Initial boiling point and boiling range | No data available         | None known              |
| Flash point                             | No data available         | None known              |
| Evaporation rate                        | No data available         | None known              |
| Flammability                            | No data available         | None known              |
| Flammability Limit in Air               |                           | None known              |
| Upper flammability or explosive limits  | No data available         |                         |
| Lower flammability or explosive limits  | No data available         |                         |
| Vapour pressure                         | No data available         | None known              |
| Relative vapour density                 | No data available         | None known              |
| Relative density                        | No data available         | None known              |
| Water solubility                        | Soluble in water          |                         |
| Solubility(ies)                         | No data available         | None known              |
| Partition coefficient                   | No data available         | None known              |
| Autoignition temperature                | No data available         | None known              |
| Decomposition temperature               |                           | None known              |
| Kinematic viscosity                     | No data available         | None known              |
| Dynamic viscosity                       | No data available         | None known              |
| Explosive properties                    | No information available. |                         |
| Oxidising properties                    | No information available. |                         |
| <u>Other information</u>                |                           |                         |
| Softening point                         | No information available  |                         |
| Molecular weight                        | No information available  |                         |
| VOC content                             | No information available  |                         |
| Liquid Density                          | No information available  |                         |
| Bulk density                            | No information available  |                         |
| Particle characteristics                | No information available  |                         |

## Section 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials None known based on information supplied.

### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

## Section 11: Toxicological information

### Acute toxicity

### Information on likely routes of exposure

#### Product Information

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms** No information available.

### Acute toxicity

.

#### Numerical measures of toxicity

No information available

The following values are calculated based on chapter 3.1 of the GHS document

**ATEmix (oral)** 33,333.30 mg/kg

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitisation** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Data used to identify the health effects** Refer to Section 16 for Key literature references and sources for data used to compile the SDS.

## Section 12: Ecological information

**Ecotoxicity**

|                                      |                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Aquatic ecotoxicity</b>           | The environmental impact of this product has not been fully investigated.                  |
| <b>Unknown aquatic toxicity</b>      | 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment. |
| <b>Terrestrial ecotoxicity</b>       | There is no data for this product.                                                         |
| <b>Persistence and degradability</b> | No information available.                                                                  |

**Bioaccumulative potential**

|                        |                                    |
|------------------------|------------------------------------|
| <b>Bioaccumulation</b> | There is no data for this product. |
|------------------------|------------------------------------|

**Mobility in soil**

|                 |                           |
|-----------------|---------------------------|
| <b>Mobility</b> | No information available. |
|-----------------|---------------------------|

**Other adverse effects**

No information available.

**Section 13: Disposal considerations****Disposal methods**

|                                            |                                   |
|--------------------------------------------|-----------------------------------|
| <b>Waste from residues/unused products</b> | Not applicable.<br>Not Hazardous. |
| <b>Contaminated packaging</b>              | Not applicable.<br>Not Hazardous. |

**Section 14: Transport information**

|                    |               |
|--------------------|---------------|
| <b><u>IATA</u></b> | Not regulated |
|--------------------|---------------|

|                    |               |
|--------------------|---------------|
| <b><u>IMDG</u></b> | Not regulated |
|--------------------|---------------|

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information

**Section 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations**

|                                                             |                  |
|-------------------------------------------------------------|------------------|
| <b>EPA New Zealand HSNO approval code or group standard</b> | To be determined |
|-------------------------------------------------------------|------------------|

|                             |                                                                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>National regulations</b> | There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDL</b>       | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AIIC</b>          | Contact supplier for inventory compliance status. |

**Legend:**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

**ENCS** - Japan Existing and New Chemical Substances

**IECSC** - China Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**Section 16: Other information**

**Revision date** 13-Feb-2024

**Revision Note** Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend** Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|                |                             |             |                                  |
|----------------|-----------------------------|-------------|----------------------------------|
| <b>TWA</b>     | TWA (time-weighted average) | <b>STEL</b> | STEL (Short Term Exposure Limit) |
| <b>Ceiling</b> | Maximum limit value         | <b>*</b>    | Skin designation                 |
| <b>C</b>       | Carcinogen                  |             |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)

U.S. Environmental Protection Agency ChemView Database

European Food Safety Authority (EFSA)

EPA (Environmental Protection Agency)

Acute Exposure Guideline Level(s) (AEGl(s))

U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act

U.S. Environmental Protection Agency High Production Volume Chemicals

Food Research Journal

Hazardous Substance Database

International Uniform Chemical Information Database (IUCLID)



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National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

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**End of Safety Data Sheet**

# SAFETY DATA SHEET

Revision date 13-Feb-2024

Revision Number 1.2

## Section 1: Identification

### Product identifier

**Product Name** Bio-Plex Pro Human Chemokine Panel 40-plex Lyophilized Control

**Catalogue Number(s)** 10032535

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Supplier

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Importer

Bio-Rad Laboratories Pty Ltd  
189 Bush Road  
Auckland  
New Zealand

**Technical Service** +64 9 415 2280 or 0508 805 500  
sales.nz@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Label elements

#### **Hazard statements**

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Other hazards which do not result in classification

No information available.

## Section 3: Composition/information on ingredients

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Non-hazardous ingredients | Proprietary | Balance  |

## Section 4: First-aid measures

### Description of first aid measures

|              |                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                                 |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids.<br>Consult a doctor. |
| Skin contact | Wash skin with soap and water.                                                                                       |
| Ingestion    | Rinse mouth.                                                                                                         |

### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

### Indication of any immediate medical attention and special treatment needed

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

### Special protective actions for fire-fighters

|                                                                |                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|

## Section 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

|                          |                                                   |
|--------------------------|---------------------------------------------------|
| Personal precautions     | Ensure adequate ventilation.                      |
| For emergency responders | Use personal protection recommended in Section 8. |

### Environmental precautions

|                           |                                                       |
|---------------------------|-------------------------------------------------------|
| Environmental precautions | See Section 12 for additional Ecological Information. |
|---------------------------|-------------------------------------------------------|

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** For long-term storage, keep the battery at temperatures between 10°C-30°C.

**Incompatible materials** None known based on information supplied.

## Section 8: Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** No special protective equipment required.

**Hand protection** No special protective equipment required.

**Skin and body protection** No special protective equipment required.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## Section 9: Physical and chemical properties

**Information on basic physical and chemical properties**

**Physical state** Solid  
**Appearance** powder or cake, lyophilised  
**Colour** white

|                                         |                           |                                |
|-----------------------------------------|---------------------------|--------------------------------|
| Odour                                   | Odourless.                |                                |
| Odour threshold                         | No information available  |                                |
| <b><u>Property</u></b>                  | <b><u>Values</u></b>      | <b><u>Remarks • Method</u></b> |
| pH                                      | 7.4                       |                                |
| Melting point / freezing point          | No data available         | None known                     |
| Initial boiling point and boiling range | No data available         | None known                     |
| Flash point                             | No data available         | None known                     |
| Evaporation rate                        | No data available         | None known                     |
| Flammability                            | No data available         | None known                     |
| Flammability Limit in Air               |                           | None known                     |
| Upper flammability or explosive limits  | No data available         |                                |
| Lower flammability or explosive limits  | No data available         |                                |
| Vapour pressure                         | No data available         | None known                     |
| Relative vapour density                 | No data available         | None known                     |
| Relative density                        | No data available         | None known                     |
| Water solubility                        | Soluble in water          |                                |
| Solubility(ies)                         | No data available         | None known                     |
| Partition coefficient                   | No data available         | None known                     |
| Autoignition temperature                | No data available         | None known                     |
| Decomposition temperature               |                           | None known                     |
| Kinematic viscosity                     | No data available         | None known                     |
| Dynamic viscosity                       | No data available         | None known                     |
| Explosive properties                    | No information available. |                                |
| Oxidising properties                    | No information available. |                                |
| <b><u>Other information</u></b>         |                           |                                |
| Softening point                         | No information available  |                                |
| Molecular weight                        | No information available  |                                |
| VOC content                             | No information available  |                                |
| Liquid Density                          | No information available  |                                |
| Bulk density                            | No information available  |                                |
| Particle characteristics                | No information available  |                                |

## Section 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials None known based on information supplied.

Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

**Section 11: Toxicological information**

Acute toxicity

Information on likely routes of exposure

**Product Information**

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. |

**Symptoms** No information available.

Acute toxicity

**Numerical measures of toxicity**  
No information available

**The following values are calculated based on chapter 3.1 of the GHS document**  
**ATEmix (oral)** 33,333.30 mg/kg

Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

**Respiratory or skin sensitisation** No information available.

**Germ cell mutagenicity** No information available.

**Carcinogenicity** No information available.

**Reproductive toxicity** No information available.

**STOT - single exposure** No information available.

**STOT - repeated exposure** No information available.

**Aspiration hazard** No information available.

**Data used to identify the health effects** Refer to Section 16 for Key literature references and sources for data used to compile the SDS.

## Section 12: Ecological information

### Ecotoxicity

|                                      |                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Aquatic ecotoxicity</b>           | The environmental impact of this product has not been fully investigated.                  |
| <b>Unknown aquatic toxicity</b>      | 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment. |
| <b>Terrestrial ecotoxicity</b>       | There is no data for this product.                                                         |
| <b>Persistence and degradability</b> | No information available.                                                                  |

### Bioaccumulative potential

|                        |                                    |
|------------------------|------------------------------------|
| <b>Bioaccumulation</b> | There is no data for this product. |
|------------------------|------------------------------------|

### Mobility in soil

|                 |                           |
|-----------------|---------------------------|
| <b>Mobility</b> | No information available. |
|-----------------|---------------------------|

### Other adverse effects

No information available.

## Section 13: Disposal considerations

### Disposal methods

|                                            |                                   |
|--------------------------------------------|-----------------------------------|
| <b>Waste from residues/unused products</b> | Not applicable.<br>Not Hazardous. |
| <b>Contaminated packaging</b>              | Not applicable.<br>Not Hazardous. |

## Section 14: Transport information

|                    |               |
|--------------------|---------------|
| <b><u>IATA</u></b> | Not regulated |
|--------------------|---------------|

|                    |               |
|--------------------|---------------|
| <b><u>IMDG</u></b> | Not regulated |
|--------------------|---------------|

**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available

**Special precautions for user**  
Please refer to the applicable dangerous goods regulations for additional information

## Section 15: Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National regulations**

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>EPA New Zealand HSNO approval code or group standard</b>                       | To be determined                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>National regulations</b>                                                       | There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Certified handlers, tracking and controlled substance license requirements</b> | <p>Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information</p> <p>Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information</p> <p>Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information</p> |

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

### International Inventories

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDSL</b>      | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AIC</b>           | Contact supplier for inventory compliance status. |

#### **Legend:**

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances  
**ENCS** - Japan Existing and New Chemical Substances  
**IECSC** - China Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances

## **Section 16: Other information**

**Revision date** 13-Feb-2024  
**Revision Note** Reformatted and updated existing information.

### **Key or legend to abbreviations and acronyms used in the safety data sheet**

#### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)



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Acute Exposure Guideline Level(s) (AEGL(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

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**End of Safety Data Sheet**

# SAFETY DATA SHEET

Revision date 13-Feb-2024

Revision Number 1.1

## Section 1: Identification

### Product identifier

**Product Name** Bio-Plex Pro Human Chemokine Detection Antibodies

**Catalogue Number(s)** 10032533

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Supplier

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Importer

Bio-Rad Laboratories Pty Ltd  
189 Bush Road  
Albany Auckland  
New Zealand

**Technical Service** +64 9 415 2280 or 0508 805 500  
sales.nz@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

|                                 |            |
|---------------------------------|------------|
| <b>Skin sensitisation</b>       | Category 1 |
| <b>Chronic aquatic toxicity</b> | Category 3 |

### Label elements



#### **Signal word**

Warning

#### **Hazard statements**

May cause an allergic skin reaction  
Harmful to aquatic life with long lasting effects

#### **Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapours/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/clothing and eye/face protection

Avoid release to the environment

#### Skin

IF ON SKIN: Wash with plenty of water and soap

If skin irritation or rash occurs: Get medical advice/attention

Take off all contaminated clothing and wash it before reuse

#### Precautionary Statements - Disposal

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

#### Other hazards which do not result in classification

Harmful to aquatic life.

Contains human source material and / or potentially infectious components

### Section 3: Composition/information on ingredients

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Trade secret              | -           | 0.01 - 0.099 |
| Trade secret              | -           | 0.001 - 0.01 |
| Non-hazardous ingredients | Proprietary | Balance      |

### Section 4: First-aid measures

#### Description of first aid measures

|                |                                                                                                                                                                                              |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General advice | Show this safety data sheet to the doctor in attendance.                                                                                                                                     |
| Inhalation     | Remove to fresh air.                                                                                                                                                                         |
| Eye contact    | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. Contains human source material and / or potentially infectious components. |
| Skin contact   | Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor. Wash skin with soap and water.                             |
| Ingestion      | Call a doctor. May present a choking hazard. Contains human source material and / or potentially infectious components.                                                                      |

#### Most important symptoms and effects, both acute and delayed

Symptoms Itching. Rashes. Hives.

Effects of Exposure No information available.

#### Indication of any immediate medical attention and special treatment needed

Note to doctors May cause sensitisation in susceptible persons. Treat symptomatically. Contains human source material and / or potentially infectious components.

### Section 5: Fire-fighting measures

#### Suitable Extinguishing Media

Suitable Extinguishing Media Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Large Fire**

CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media**

Do not scatter spilled material with high pressure water streams.

**Specific hazards arising from the chemical**

**Specific hazards arising from the chemical**

Product is or contains a sensitiser. May cause sensitisation by skin contact.

**Special protective actions for fire-fighters**

**Special protective equipment and precautions for fire-fighters**

Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.

## Section 6: Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions**

Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**For emergency responders**

Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions**

See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment**

Do not allow into any sewer, on the ground or into any body of water.

**Methods for cleaning up**

Clean contaminated surface thoroughly. This product should be considered as hazardous if the battery is compromised and starts to leak. Collect spillage with a shovel and broom, or similar. Use: Disinfectant.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

**Precautions for safe handling**

**Advice on safe handling**

Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.

**General hygiene considerations**

Follow universal and standard precautions for handling potentially infectious materials.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. For long-term storage, keep the battery at temperatures between 10°C-30°C.

**Incompatible materials**

Metals.

## Section 8: Exposure controls/personal protection

### Control parameters

#### Exposure Limits

| Chemical name | New Zealand                                          | Australia                                     | ACGIH TLV                                                                                    | United Kingdom                                                   |
|---------------|------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Trade secret  | Ceiling: 0.11 ppm<br>Ceiling: 0.29 mg/m <sup>3</sup> | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup><br>Sodium azide<br>Ceiling: 0.11 ppm<br>Hydrazoic acid vapor | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.3 mg/m <sup>3</sup><br>Sk* |

#### Biological occupational exposure limits

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

### Appropriate engineering controls

#### Engineering controls

Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

#### Eye/face protection

Wear safety glasses with side shields (or goggles).

#### Hand protection

Wear suitable gloves.

#### Skin and body protection

Wear suitable protective clothing.

#### Respiratory protection

No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

#### Environmental exposure controls

No information available.

## Section 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                 |                          |
|-----------------|--------------------------|
| Physical state  | Liquid                   |
| Appearance      | aqueous solution         |
| Colour          | colourless               |
| Odour           | Odourless.               |
| Odour threshold | No information available |

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7.4               |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | > 100 °C          |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |
| Lower flammability or explosive limits  | No data available |                         |
| Vapour pressure                         | No data available | None known              |
| Relative vapour density                 | No data available | None known              |
| Relative density                        | No data available | None known              |
| Water solubility                        | Miscible in water |                         |
| Solubility(ies)                         | No data available | None known              |
| Partition coefficient                   | No data available | None known              |

|                           |                           |            |
|---------------------------|---------------------------|------------|
| Autoignition temperature  | No data available         | None known |
| Decomposition temperature |                           | None known |
| Kinematic viscosity       | No data available         | None known |
| Dynamic viscosity         | No data available         | None known |
| Explosive properties      | No information available. |            |
| Oxidising properties      | No information available. |            |
| <u>Other information</u>  |                           |            |
| Softening point           | No information available  |            |
| Molecular weight          | No information available  |            |
| VOC content               | No information available  |            |
| Liquid Density            | No information available  |            |
| Bulk density              | No information available  |            |
| Particle characteristics  | No information available  |            |

## Section 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials Metals.

### Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

## Section 11: Toxicological information

### Acute toxicity

### Information on likely routes of exposure

#### Product Information

Inhalation Specific test data for the substance or mixture is not available.

Eye contact Specific test data for the substance or mixture is not available.

Skin contact May cause sensitisation by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons (based on components).

|                                       |                                                                   |
|---------------------------------------|-------------------------------------------------------------------|
| <b>Ingestion</b>                      | Specific test data for the substance or mixture is not available. |
| <b>Symptoms</b>                       | Itching. Rashes. Hives.                                           |
| <b>Acute toxicity</b>                 | .                                                                 |
| <b>Numerical measures of toxicity</b> |                                                                   |

#### Component Information

| Chemical name | Oral LD50          | Dermal LD50              | Inhalation LC50               |
|---------------|--------------------|--------------------------|-------------------------------|
| Trade secret  | = 27 mg/kg ( Rat ) | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |
| Trade secret  | = 53 mg/kg ( Rat ) | = 87.12 mg/kg ( Rabbit ) | -                             |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                                 |                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>                | No information available.                                                                       |
| <b>Serious eye damage/eye irritation</b>        | No information available.                                                                       |
| <b>Respiratory or skin sensitisation</b>        | May cause an allergic skin reaction.                                                            |
| <b>Germ cell mutagenicity</b>                   | No information available.                                                                       |
| <b>Carcinogenicity</b>                          | No information available.                                                                       |
| <b>Reproductive toxicity</b>                    | No information available.                                                                       |
| <b>STOT - single exposure</b>                   | No information available.                                                                       |
| <b>STOT - repeated exposure</b>                 | No information available.                                                                       |
| <b>Aspiration hazard</b>                        | No information available.                                                                       |
| <b>Data used to identify the health effects</b> | Refer to Section 16 for Key literature references and sources for data used to compile the SDS. |

## Section 12: Ecological information

#### Ecotoxicity

|                                 |                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Aquatic ecotoxicity</b>      | Harmful to aquatic life with long lasting effects.                                                |
| <b>Unknown aquatic toxicity</b> | 13.02609 % of the mixture consists of component(s) of unknown hazards to the aquatic environment. |

| Chemical name | Algae/aquatic plants | Fish                                                                 | Crustacea |
|---------------|----------------------|----------------------------------------------------------------------|-----------|
| Trade secret  | -                    | LC50: =0.8mg/L (96h,<br>Oncorhynchus mykiss)<br>LC50: =0.7mg/L (96h, | -         |

|  |  |                                                                      |  |
|--|--|----------------------------------------------------------------------|--|
|  |  | Lepomis macrochirus<br>LC50: =5.46mg/L (96h,<br>Pimephales promelas) |  |
|--|--|----------------------------------------------------------------------|--|

**Terrestrial ecotoxicity** There is no data for this product.

**Persistence and degradability** No information available.

### Bioaccumulative potential

#### Bioaccumulation

##### Component Information

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | 0.7                   |

### Mobility in soil

**Mobility** No information available.

### Other adverse effects

No information available.

## Section 13: Disposal considerations

### Disposal methods

#### Waste from residues/unused products

Dispose of product in packaging in a way that is consistent with the EPA Consolidation 30 April 2021 of the Hazardous Substances (Disposal) Notice 2017 and the Act.

Treat the substance using a method that changes the characteristics or composition of the substance so that the substance is no longer a hazardous substance; or export the substance from New Zealand as waste.

Substances which are hazardous to human health or corrosive to metals – may be discharged into the environment if a tolerable exposure limit has been set for the substance (or a component of that substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the tolerable exposure limit. If there is no tolerable exposure limit for the substance, then it may only be discharged into the environment if the substance is very rapidly converted to substances that are not hazardous substances.

Environmentally hazardous substances – if the substance, or if it contains a component that is hazardous to the aquatic environment or bioaccumulative and not rapidly degradable, then any component that is bioaccumulative and not rapidly degradable must be removed. The product may only be discharged into the environment if an environmental exposure limit has been set for the substance (or a component of the substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the environmental exposure limit.

#### Contaminated packaging

For packages that have been in direct contact with hazardous substances, the person must ensure that the package is rendered incapable of containing any substance. It must be disposed of in a manner that is consistent with the requirements for disposal of the substance that it contained, taking into account the material the package is manufactured from.

Packages may only be reused or recycled if:

- the substance has a physical hazard other than corrosive to metal, and has been treated to remove any residual contents of the hazardous substance;
- or for substances that have a health or environmental hazard, or corrosive to metal, the contents of the residue in the package are below the threshold for the substance to be



classified as hazardous in the Hazardous Substances (Hazard Classification) Notice 2020.

## Section 14: Transport information

**IATA** Not regulated

**IMDG** Not regulated

### Transport in bulk according to Annex II of MARPOL and the IBC Code

No information available

### Special precautions for user

Please refer to the applicable dangerous goods regulations for additional information

## Section 15: Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

**EPA New Zealand HSNO approval code or group standard** To be determined

**National regulations** There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements** Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information  
Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information  
Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

#### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### International Inventories

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDSL</b>      | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AIIC</b>          | Contact supplier for inventory compliance status. |

#### **Legend:**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

**ENCS** - Japan Existing and New Chemical Substances  
**IECSC** - China Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances

## Section 16: Other information

**Revision date** 13-Feb-2024

**Revision Note** Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend** Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AEGl(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals  
Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

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**End of Safety Data Sheet**

# SAFETY DATA SHEET

Revision date 22-Aug-2023

Revision Number 1

## Section 1: Identification

### Product identifier

**Product Name** Bio-Plex Pro Human Chemokine 40-Plex Beads

**Catalogue Number(s)** 10032444

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Supplier

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Importer

Bio-Rad Laboratories Pty Ltd  
189 Bush Road  
Albany Auckland  
New Zealand

**Technical Service** +64 9 415 2280 or 0508 805 500  
sales.nz@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

|                                 |            |
|---------------------------------|------------|
| <b>Skin sensitisation</b>       | Category 1 |
| <b>Chronic aquatic toxicity</b> | Category 3 |

### Label elements



#### **Signal word**

Warning

#### **Hazard statements**

May cause an allergic skin reaction  
Harmful to aquatic life with long lasting effects

#### **Precautionary Statements - Prevention**

Avoid breathing dust/fume/gas/mist/vapours/spray  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/clothing and eye/face protection

Avoid release to the environment

**Skin**

IF ON SKIN: Wash with plenty of water and soap

If skin irritation or rash occurs: Get medical advice/attention

Take off all contaminated clothing and wash it before reuse

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

Harmful to aquatic life. Contains animal source material. (Cattle).

### Section 3: Composition/information on ingredients

| Chemical name             | CAS No      | Weight-%     |
|---------------------------|-------------|--------------|
| Water                     | 7732-18-5   | 50 - 100     |
| Magnetic Beads            | NO-CAS-23   | 20 - 35      |
| Trade secret              | -           | 10 - 20      |
| Trade secret              | -           | 5 - 10       |
| Trade secret              | -           | 5 - 10       |
| Trade secret              | -           | 1 - 2.5      |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Trade secret              | -           | 0.1 - 0.299  |
| Antibodies                | NO-CAS-81   | 0.1 - 0.299  |
| Trade secret              | -           | 0.001 - 0.01 |
| Trade secret              | -           | 0.001 - 0.01 |
| Non-hazardous ingredients | Proprietary | Balance      |

### Section 4: First-aid measures

**Description of first aid measures**

**General advice**

Show this safety data sheet to the doctor in attendance.

**Inhalation**

Remove to fresh air.

**Eye contact**

Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor.

**Skin contact**

Wash with soap and water. May cause an allergic skin reaction. In the case of skin irritation or allergic reactions see a doctor.

**Ingestion**

Rinse mouth.

**Most important symptoms and effects, both acute and delayed**

**Symptoms**

Itching. Rashes. Hives.

**Effects of Exposure**

No information available.

**Indication of any immediate medical attention and special treatment needed**

**Note to doctors**

May cause sensitisation in susceptible persons. Treat symptomatically.

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

### Specific hazards arising from the chemical

**Specific hazards arising from the chemical** Product is or contains a sensitiser. May cause sensitisation by skin contact.

### Special protective actions for fire-fighters

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.

## Section 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**For emergency responders** Use personal protection recommended in Section 8.

### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

### Precautions to prevent secondary hazards

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Ensure adequate ventilation. In case of insufficient ventilation, wear suitable respiratory equipment. Do not eat, drink or smoke when using this product. Take off contaminated clothing and wash it before reuse.

### Conditions for safe storage, including any incompatibilities

**Storage Conditions** Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

## Section 8: Exposure controls/personal protection

### Control parameters

#### Exposure Limits

| Chemical name | New Zealand                                          | Australia                                     | ACGIH TLV                                                                                    | United Kingdom                                                   |
|---------------|------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Trade secret  | TWA: 10 mg/m <sup>3</sup>                            | TWA: 10 mg/m <sup>3</sup>                     | TWA: 10 mg/m <sup>3</sup>                                                                    | TWA: 10 mg/m <sup>3</sup><br>STEL: 20 mg/m <sup>3</sup>          |
| Trade secret  | Ceiling: 2 mg/m <sup>3</sup>                         | Peak: 2 mg/m <sup>3</sup>                     | Ceiling: 2 mg/m <sup>3</sup>                                                                 | STEL: 2 mg/m <sup>3</sup>                                        |
| Trade secret  | Ceiling: 0.11 ppm<br>Ceiling: 0.29 mg/m <sup>3</sup> | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup><br>Sodium azide<br>Ceiling: 0.11 ppm<br>Hydrazoic acid vapor | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.3 mg/m <sup>3</sup><br>Sk* |

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## Section 9: Physical and chemical properties

### Information on basic physical and chemical properties

**Physical state** Liquid  
**Appearance** Dilute bead suspension in aqueous solution  
**Colour** Cloudy  
**Odour** Odourless.  
**Odour threshold** No information available

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 7.4               |                         |
| Melting point / freezing point          | No data available | None known              |
| Initial boiling point and boiling range | > 100 °C          |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |
| Flammability                            | No data available | None known              |
| Flammability Limit in Air               |                   | None known              |
| Upper flammability or explosive limits  | No data available |                         |

|                                               |                           |            |
|-----------------------------------------------|---------------------------|------------|
| <b>Lower flammability or explosive limits</b> | No data available         |            |
| <b>Vapour pressure</b>                        | No data available         | None known |
| <b>Relative vapour density</b>                | No data available         | None known |
| <b>Relative density</b>                       | No data available         | None known |
| <b>Water solubility</b>                       | Partially miscible        |            |
| <b>Solubility(ies)</b>                        | No data available         | None known |
| <b>Partition coefficient</b>                  | No data available         | None known |
| <b>Autoignition temperature</b>               | No data available         | None known |
| <b>Decomposition temperature</b>              |                           | None known |
| <b>Kinematic viscosity</b>                    | No data available         | None known |
| <b>Dynamic viscosity</b>                      | No data available         | None known |
| <b>Explosive properties</b>                   | No information available. |            |
| <b>Oxidising properties</b>                   | No information available. |            |
| <b><u>Other information</u></b>               |                           |            |
| <b>Softening point</b>                        | No information available  |            |
| <b>Molecular weight</b>                       | No information available  |            |
| <b>VOC content</b>                            | No information available  |            |
| <b>Liquid Density</b>                         | No information available  |            |
| <b>Bulk density</b>                           | No information available  |            |
| <b>Particle characteristics</b>               | No information available  |            |

## Section 10: Stability and reactivity

### Reactivity

**Reactivity** No information available.

### Chemical stability

**Stability** Stable under normal conditions.

### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

### Possibility of hazardous reactions

**Possibility of hazardous reactions** None under normal processing.

### Conditions to avoid

**Conditions to avoid** None known based on information supplied.

### Incompatible materials

**Incompatible materials** None known based on information supplied.

### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

## Section 11: Toxicological information

### Acute toxicity

### Information on likely routes of exposure

**Product Information**

|                                       |                                                                                                                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                     | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| <b>Eye contact</b>                    | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| <b>Skin contact</b>                   | May cause sensitisation by skin contact. Specific test data for the substance or mixture is not available. Repeated or prolonged skin contact may cause allergic reactions with susceptible persons. (based on components). |
| <b>Ingestion</b>                      | Specific test data for the substance or mixture is not available.                                                                                                                                                           |
| <b>Symptoms</b>                       | Itching. Rashes. Hives.                                                                                                                                                                                                     |
| <b>Acute toxicity</b>                 | .                                                                                                                                                                                                                           |
| <b>Numerical measures of toxicity</b> |                                                                                                                                                                                                                             |

The following values are calculated based on chapter 3.1 of the GHS document  
ATEmix (oral) 50,476.20 mg/kg

#### Component Information

| Chemical name | Oral LD50             | Dermal LD50              | Inhalation LC50               |
|---------------|-----------------------|--------------------------|-------------------------------|
| Water         | > 90 mL/kg ( Rat )    | -                        | -                             |
| Trade secret  | = 29700 mg/kg ( Rat ) | -                        | -                             |
| Trade secret  | = 3 g/kg ( Rat )      | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h         |
| Trade secret  | = 15900 mg/kg ( Rat ) | -                        | -                             |
| Trade secret  | = 325 mg/kg ( Rat )   | = 1350 mg/kg ( Rabbit )  | -                             |
| Trade secret  | = 27 mg/kg ( Rat )    | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |
| Trade secret  | = 53 mg/kg ( Rat )    | = 87.12 mg/kg ( Rabbit ) | -                             |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                                          |                                      |
|------------------------------------------|--------------------------------------|
| <b>Skin corrosion/irritation</b>         | No information available.            |
| <b>Serious eye damage/eye irritation</b> | No information available.            |
| <b>Respiratory or skin sensitisation</b> | May cause an allergic skin reaction. |
| <b>Germ cell mutagenicity</b>            | No information available.            |
| <b>Carcinogenicity</b>                   | No information available.            |
| <b>Reproductive toxicity</b>             | No information available.            |
| <b>STOT - single exposure</b>            | No information available.            |
| <b>STOT - repeated exposure</b>          | No information available.            |



|                                                 |                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Aspiration hazard</b>                        | No information available.                                                                       |
| <b>Data used to identify the health effects</b> | Refer to Section 16 for Key literature references and sources for data used to compile the SDS. |

## Section 12: Ecological information

### Ecotoxicity

**Aquatic ecotoxicity** Harmful to aquatic life with long lasting effects.

**Unknown aquatic toxicity** 0 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                                                                      | Crustacea                                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> ) | EC50: =1000mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> ) |
| Trade secret  | -                    | LC50: =45.4mg/L (96h, <i>Oncorhynchus mykiss</i> )                                                                                                                                                                                                                                                                                                        | -                                                                                                    |
| Trade secret  | -                    | LC50: =0.8mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =0.7mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =5.46mg/L (96h, <i>Pimephales promelas</i> )                                                                                                                                                                                              | -                                                                                                    |

**Terrestrial ecotoxicity** There is no data for this product.

| Chemical name | Earthworm                                                                                     | Avian | Honeybees |
|---------------|-----------------------------------------------------------------------------------------------|-------|-----------|
| Trade secret  | Acute Toxicity: LC50 0.1 - 1 mg/cm <sup>2</sup> ( <i>Eisenia foetida</i> , 48 h filter paper) | -     | -         |

**Persistence and degradability** No information available.

### Bioaccumulative potential

#### Bioaccumulation

#### Component Information

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | 0.7                   |

### Mobility in soil

**Mobility** No information available.

### Other adverse effects

No information available.

## Section 13: Disposal considerations

### Disposal methods

#### **Waste from residues/unused products**

Dispose of product in packaging in a way that is consistent with the EPA Consolidation 30 April 2021 of the Hazardous Substances (Disposal) Notice 2017 and the Act.

Treat the substance using a method that changes the characteristics or composition of the substance so that the substance is no longer a hazardous substance; or export the substance from New Zealand as waste.

Substances which are hazardous to human health or corrosive to metals – may be discharged into the environment if a tolerable exposure limit has been set for the substance (or a component of that substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the tolerable exposure limit. If there is no tolerable exposure limit for the substance, then it may only be discharged into the environment if the substance is very rapidly converted to substances that are not hazardous substances.

Environmentally hazardous substances – if the substance, or if it contains a component that is hazardous to the aquatic environment or bioaccumulative and not rapidly degradable, then any component that is bioaccumulative and not rapidly degradable must be removed. The product may only be discharged into the environment if an environmental exposure limit has been set for the substance (or a component of the substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the environmental exposure limit.

Dispose of in accordance with local regulations.

Dispose of waste in accordance with environmental legislation.

#### **Contaminated packaging**

For packages that have been in direct contact with hazardous substances, the person must ensure that the package is rendered incapable of containing any substance. It must be disposed of in a manner that is consistent with the requirements for disposal of the substance that it contained, taking into account the material the package is manufactured from.

Packages may only be reused or recycled if:

- the substance has a physical hazard other than corrosive to metal, and has been treated to remove any residual contents of the hazardous substance;
- or for substances that have a health or environmental hazard, or corrosive to metal, the contents of the residue in the package are below the threshold for the substance to be classified as hazardous in the Hazardous Substances (Hazard Classification) Notice 2020.

## Section 14: Transport information

### IATA

Not regulated

### IMDG

Not regulated

#### **Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

#### **Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information

## Section 15: Regulatory information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National regulations

EPA New Zealand HSNO approval code or group standard To be determined

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>National regulations</b>                                                       | There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Certified handlers, tracking and controlled substance license requirements</b> | <p>Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information</p> <p>Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information</p> <p>Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information</p> |

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDSL</b>      | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AIIC</b>          | Contact supplier for inventory compliance status. |

**Legend:**

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances  
**ENCS** - Japan Existing and New Chemical Substances  
**IECSC** - China Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances

**Section 16: Other information**

**Revision date** 22-Aug-2023  
**Revision Note** Significant changes throughout SDS. Review all sections.

**Key or legend to abbreviations and acronyms used in the safety data sheet****Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|                |                             |             |                                  |
|----------------|-----------------------------|-------------|----------------------------------|
| <b>TWA</b>     | TWA (time-weighted average) | <b>STEL</b> | STEL (Short Term Exposure Limit) |
| <b>Ceiling</b> | Maximum limit value         | <b>*</b>    | Skin designation                 |
| <b>C</b>       | Carcinogen                  |             |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
U.S. Environmental Protection Agency ChemView Database  
European Food Safety Authority (EFSA)  
EPA (Environmental Protection Agency)  
Acute Exposure Guideline Level(s) (AELG(s))  
U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
U.S. Environmental Protection Agency High Production Volume Chemicals

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Food Research Journal  
Hazardous Substance Database  
International Uniform Chemical Information Database (IUCLID)  
National Institute of Technology and Evaluation (NITE)  
Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

**End of Safety Data Sheet**

# SAFETY DATA SHEET

Revision date 04-Oct-2023

Revision Number 1.2

## Section 1: Identification

### Product identifier

**Product Name** Bio-Plex Detection Antibody Diluent HB

**Catalogue Number(s)** 10032400, 10031831, 12005852

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals

**Uses advised against** No information available

### Details of the supplier of the safety data sheet

#### Supplier

Bio-Rad Laboratories Inc.  
1000 Alfred Nobel Drive  
Hercules, CA 94547  
USA

#### Manufacturer

Bio-Rad Laboratories, Life Science Group  
2000 Alfred Nobel Drive  
Hercules, California 94547  
USA

#### Importer

Bio-Rad Laboratories Pty Ltd  
189 Bush Road  
Auckland  
New Zealand

**Technical Service** +64 9 415 2280 or 0508 805 500  
sales.nz@bio-rad.com

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Label elements

#### **Hazard statements**

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Other hazards which do not result in classification

Contains animal source material. (Cattle).

## Section 3: Composition/information on ingredients

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name | CAS No    | Weight-% |
|---------------|-----------|----------|
| Water         | 7732-18-5 | 50 - 100 |
| Trade secret  | -         | 5 - 10   |

| Chemical name | CAS No    | Weight-%     |
|---------------|-----------|--------------|
| Antibodies    | NO-CAS-81 | 1 - 2.5      |
| Trade secret  | -         | 0.3 - 0.99   |
| Trade secret  | -         | 0.1 - 0.299  |
| Trade secret  | -         | 0.1 - 0.299  |
| Trade secret  | -         | 0.01 - 0.099 |
| Trade secret  | -         | 0.01 - 0.099 |
| Trade secret  | -         | 0.001 - 0.01 |
| Trade secret  | -         | < 0.001      |
| Trade secret  | -         | < 0.001      |

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Non-hazardous ingredients | Proprietary | Balance  |

## Section 4: First-aid measures

### Description of first aid measures

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                              |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| Skin contact | Wash skin with soap and water.                                                                                    |
| Ingestion    | Rinse mouth.                                                                                                      |

### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

### Indication of any immediate medical attention and special treatment needed

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

### Special protective actions for fire-fighters

|                                                                |                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|

## Section 6: Accidental release measures

**Personal precautions, protective equipment and emergency procedures**

**Personal precautions** Ensure adequate ventilation.

**For emergency responders** Use personal protection recommended in Section 8.

**Environmental precautions**

**Environmental precautions** See Section 12 for additional Ecological Information.

**Methods and material for containment and cleaning up**

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

**Precautions to prevent secondary hazards**

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

**Precautions for safe handling**

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** For long-term storage, keep the battery at temperatures between 10°C-30°C. Store according to product and label instructions.

**Incompatible materials** Metals.

## Section 8: Exposure controls/personal protection

**Control parameters**

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | New Zealand                                          | Australia                                     | ACGIH TLV                                                                                    | United Kingdom                                                   |
|---------------|------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Trade secret  | Ceiling: 0.11 ppm<br>Ceiling: 0.29 mg/m <sup>3</sup> | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup><br>Sodium azide<br>Ceiling: 0.11 ppm<br>Hydrazoic acid vapor | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.3 mg/m <sup>3</sup><br>Sk* |

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

|                                        |                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face protection</b>             | No special protective equipment required.                                                                                                                                |
| <b>Hand protection</b>                 | No special protective equipment required.                                                                                                                                |
| <b>Skin and body protection</b>        | No special protective equipment required.                                                                                                                                |
| <b>Respiratory protection</b>          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                |

## Section 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                                | <u>Values</u>             | <u>Remarks • Method</u> |
|------------------------------------------------|---------------------------|-------------------------|
| <b>pH</b>                                      | 7.4                       |                         |
| <b>Melting point / freezing point</b>          | No data available         | None known              |
| <b>Initial boiling point and boiling range</b> | 100 °C                    |                         |
| <b>Flash point</b>                             | No data available         | None known              |
| <b>Evaporation rate</b>                        | No data available         | None known              |
| <b>Flammability</b>                            | No data available         | None known              |
| <b>Flammability Limit in Air</b>               |                           | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available         |                         |
| <b>Lower flammability or explosive limits</b>  | No data available         |                         |
| <b>Vapour pressure</b>                         | No data available         | None known              |
| <b>Relative vapour density</b>                 | No data available         | None known              |
| <b>Relative density</b>                        | No data available         | None known              |
| <b>Water solubility</b>                        | Miscible in water         |                         |
| <b>Solubility(ies)</b>                         | No data available         | None known              |
| <b>Partition coefficient</b>                   | No data available         | None known              |
| <b>Autoignition temperature</b>                | No data available         | None known              |
| <b>Decomposition temperature</b>               |                           | None known              |
| <b>Kinematic viscosity</b>                     | No data available         | None known              |
| <b>Dynamic viscosity</b>                       | No data available         | None known              |
| <b>Explosive properties</b>                    | No information available. |                         |
| <b>Oxidising properties</b>                    | No information available. |                         |

### Other information

|                                 |                          |
|---------------------------------|--------------------------|
| <b>Softening point</b>          | No information available |
| <b>Molecular weight</b>         | No information available |
| <b>VOC content</b>              | No information available |
| <b>Liquid Density</b>           | No information available |
| <b>Bulk density</b>             | No information available |
| <b>Particle characteristics</b> | No information available |

## Section 10: Stability and reactivity

### Reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

### Chemical stability

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|



**Explosion data**

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

**Possibility of hazardous reactions**

**Possibility of hazardous reactions** Avoid contact with metals. This product contains Sodium azide. Sodium azide can react with Copper, Brass, Lead, and solder in piping systems to form explosive compounds and toxic gases.

**Conditions to avoid**

Conditions to avoid None known based on information supplied.

**Incompatible materials**

Incompatible materials Metals.

**Hazardous decomposition products**

Hazardous decomposition products None known based on information supplied.

|                                              |
|----------------------------------------------|
| <b>Section 11: Toxicological information</b> |
|----------------------------------------------|

**Acute toxicity****Information on likely routes of exposure****Product Information**

Inhalation Specific test data for the substance or mixture is not available.

Eye contact Specific test data for the substance or mixture is not available.

Skin contact Specific test data for the substance or mixture is not available.

Ingestion Specific test data for the substance or mixture is not available.

Symptoms No information available.

**Acute toxicity**

.

**Numerical measures of toxicity**

No information available

| Chemical name | Oral LD50             | Dermal LD50              | Inhalation LC50               |
|---------------|-----------------------|--------------------------|-------------------------------|
| Water         | > 90 mL/kg ( Rat )    | -                        | -                             |
| Trade secret  | = 3 g/kg ( Rat )      | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h         |
| Trade secret  | = 8290 mg/kg ( Rat )  | > 7940 mg/kg ( Rabbit )  | > 0.83 mg/L ( Rat ) 4 h       |
| Trade secret  | = 27 mg/kg ( Rat )    | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |
| Trade secret  | = 37000 mg/kg ( Rat ) | -                        | > 5.1 mg/L ( Rat ) 4 h        |
| Trade secret  | = 53 mg/kg ( Rat )    | = 87.12 mg/kg ( Rabbit ) | -                             |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                                 |                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>                | No information available.                                                                       |
| <b>Serious eye damage/eye irritation</b>        | No information available.                                                                       |
| <b>Respiratory or skin sensitisation</b>        | No information available.                                                                       |
| <b>Germ cell mutagenicity</b>                   | No information available.                                                                       |
| <b>Carcinogenicity</b>                          | No information available.                                                                       |
| <b>Reproductive toxicity</b>                    | No information available.                                                                       |
| <b>STOT - single exposure</b>                   | No information available.                                                                       |
| <b>STOT - repeated exposure</b>                 | No information available.                                                                       |
| <b>Aspiration hazard</b>                        | No information available.                                                                       |
| <b>Data used to identify the health effects</b> | Refer to Section 16 for Key literature references and sources for data used to compile the SDS. |

## Section 12: Ecological information

### Ecotoxicity

**Aquatic ecotoxicity** The environmental impact of this product has not been fully investigated.

**Unknown aquatic toxicity** 0.095 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                                                                      | Crustacea                                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =12946mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: 6020 - 7070mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: =7050mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 6420 - 6700mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 4747 - 7824mg/L (96h, <i>Oncorhynchus mykiss</i> ) | EC50: =1000mg/L (48h, <i>Daphnia magna</i> )<br>EC50: 340.7 - 469.2mg/L (48h, <i>Daphnia magna</i> ) |
| Trade secret  | -                    | LC50: =0.8mg/L (96h, <i>Oncorhynchus mykiss</i> )<br>LC50: =0.7mg/L (96h, <i>Lepomis macrochirus</i> )<br>LC50: =5.46mg/L (96h, <i>Pimephales promelas</i> )                                                                                                                                                                                              | -                                                                                                    |

**Terrestrial ecotoxicity** There is no data for this product.

| Chemical name | Earthworm                                                                | Avian | Honeybees |
|---------------|--------------------------------------------------------------------------|-------|-----------|
| Trade secret  | Acute Toxicity: LC50 0.1 - 1 mg/cm2 (Eisenia foetida, 48 h filter paper) | -     | -         |

**Persistence and degradability** No information available.

#### Bioaccumulative potential

**Bioaccumulation** There is no data for this product.

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | 0.7                   |

#### Mobility in soil

**Mobility** No information available.

#### Other adverse effects

No information available.

### Section 13: Disposal considerations

#### Disposal methods

**Waste from residues/unused products** Not applicable.  
Not Hazardous.  
Dispose of in accordance with local regulations.  
Dispose of waste in accordance with environmental legislation.  
Flush pipes with water frequently if discarding solutions containing Sodium azide into metal piping systems.

**Contaminated packaging** Not applicable.  
Not Hazardous.

### Section 14: Transport information

**IATA** Not regulated

**IMDG** Not regulated

#### **Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

#### **Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information

### Section 15: Regulatory information

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

##### National regulations

**EPA New Zealand HSNO approval code or group standard** To be determined

**National regulations** There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

**International Inventories**

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDL</b>       | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AIIC</b>          | Contact supplier for inventory compliance status. |

**Legend:**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

**ENCS** - Japan Existing and New Chemical Substances

**IECSC** - China Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**Section 16: Other information**

**Revision date** 04-Oct-2023

**Revision Note** Reformatted and updated existing information.

**Key or legend to abbreviations and acronyms used in the safety data sheet**

Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

**Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)

U.S. Environmental Protection Agency ChemView Database

European Food Safety Authority (EFSA)

EPA (Environmental Protection Agency)

Acute Exposure Guideline Level(s) (AEGL(s))

U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act

U.S. Environmental Protection Agency High Production Volume Chemicals

Food Research Journal

Hazardous Substance Database

International Uniform Chemical Information Database (IUCLID)

National Institute of Technology and Evaluation (NITE)

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Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
NIOSH (National Institute for Occupational Safety and Health)  
National Library of Medicine's ChemID Plus (NLM CIP)  
National Library of Medicine's PubMed database (NLM PUBMED)  
National Toxicology Program (NTP)  
New Zealand's Chemical Classification and Information Database (CCID)  
Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
Organisation for Economic Co-operation and Development Screening Information Data Set  
World Health Organization

**Disclaimer**

**The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text**

**End of Safety Data Sheet**

# SAFETY DATA SHEET

Revision date 04-Oct-2023

Revision Number 1.2

## Section 1: Identification

### Product identifier

**Product Name** Bio-Plex Sample Diluent  
**Catalogue Number(s)** 10014641, 9704423, 9703895, 171305043, 10022628, 10041561, 12005851

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of the supplier of the safety data sheet

|                                                                                                      |                                                                                                                                 |                                                                                                    |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Supplier</b><br>Bio-Rad Laboratories Inc.<br>1000 Alfred Nobel Drive<br>Hercules, CA 94547<br>USA | <b>Manufacturer</b><br>Bio-Rad Laboratories, Life Science Group<br>2000 Alfred Nobel Drive<br>Hercules, California 94547<br>USA | <b>Importer</b><br>Bio-Rad Laboratories Pty Ltd<br>189 Bush Road<br>Albany Auckland<br>New Zealand |
| <b>Technical Service</b>                                                                             | +64 9 415 2280 or 0508 805 500<br>sales.nz@bio-rad.com                                                                          |                                                                                                    |

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Label elements

#### **Hazard statements**

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Other hazards which do not result in classification

Contains animal source material. (Cattle).

## Section 3: Composition/information on ingredients

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name | CAS No    | Weight-%   |
|---------------|-----------|------------|
| Water         | 7732-18-5 | 50 - 100   |
| Trade secret  | -         | 0.3 - 0.99 |

| Chemical name | CAS No    | Weight-%     |
|---------------|-----------|--------------|
| HBR 1 Blocker | NO-CAS-18 | 0.1 - 0.299  |
| Trade secret  | -         | 0.1 - 0.299  |
| Trade secret  | -         | 0.1 - 0.299  |
| Trade secret  | -         | 0.1 - 0.299  |
| Trade secret  | -         | 0.1 - 0.299  |
| Trade secret  | -         | 0.01 - 0.099 |
| Trade secret  | -         | 0.01 - 0.099 |

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Non-hazardous ingredients | Proprietary | Balance  |

## Section 4: First-aid measures

### Description of first aid measures

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                              |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| Skin contact | Wash skin with soap and water.                                                                                    |
| Ingestion    | Rinse mouth.                                                                                                      |

### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

### Indication of any immediate medical attention and special treatment needed

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

### Special protective actions for fire-fighters

|                                                                |                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|

## Section 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Ensure adequate ventilation.

**For emergency responders** Use personal protection recommended in Section 8.

#### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

#### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

#### Precautions to prevent secondary hazards

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

#### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

#### Conditions for safe storage, including any incompatibilities

**Storage Conditions** For long-term storage, keep the battery at temperatures between 10°C-30°C. Store according to product and label instructions.

**Incompatible materials** Metals.

## Section 8: Exposure controls/personal protection

#### Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | New Zealand                                          | Australia                                     | ACGIH TLV                                                                                    | United Kingdom                                                   |
|---------------|------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Trade secret  | Ceiling: 0.11 ppm<br>Ceiling: 0.29 mg/m <sup>3</sup> | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup><br>Sodium azide<br>Ceiling: 0.11 ppm<br>Hydrazoic acid vapor | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.3 mg/m <sup>3</sup><br>Sk* |

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

#### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

#### Individual protection measures, such as personal protective equipment

**Eye/face protection** No special protective equipment required.



|                                        |                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hand protection</b>                 | No special protective equipment required.                                                                                                                                |
| <b>Skin and body protection</b>        | No special protective equipment required.                                                                                                                                |
| <b>Respiratory protection</b>          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                |

## Section 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                                | <u>Values</u>             | <u>Remarks • Method</u> |
|------------------------------------------------|---------------------------|-------------------------|
| <b>pH</b>                                      | 7.4                       |                         |
| <b>Melting point / freezing point</b>          | 0 °C                      |                         |
| <b>Initial boiling point and boiling range</b> | 100 °C                    |                         |
| <b>Flash point</b>                             | No data available         | None known              |
| <b>Evaporation rate</b>                        | No data available         | None known              |
| <b>Flammability</b>                            | No data available         | None known              |
| <b>Flammability Limit in Air</b>               |                           | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available         |                         |
| <b>Lower flammability or explosive limits</b>  | No data available         |                         |
| <b>Vapour pressure</b>                         | No data available         | None known              |
| <b>Relative vapour density</b>                 | No data available         | None known              |
| <b>Relative density</b>                        | No data available         | None known              |
| <b>Water solubility</b>                        | Miscible in water         |                         |
| <b>Solubility(ies)</b>                         | No data available         | None known              |
| <b>Partition coefficient</b>                   | No data available         | None known              |
| <b>Autoignition temperature</b>                | No data available         | None known              |
| <b>Decomposition temperature</b>               |                           | None known              |
| <b>Kinematic viscosity</b>                     | No data available         | None known              |
| <b>Dynamic viscosity</b>                       | No data available         | None known              |
| <b>Explosive properties</b>                    | No information available. |                         |
| <b>Oxidising properties</b>                    | No information available. |                         |

### Other information

|                                 |                          |
|---------------------------------|--------------------------|
| <b>Softening point</b>          | No information available |
| <b>Molecular weight</b>         | No information available |
| <b>VOC content</b>              | No information available |
| <b>Liquid Density</b>           | No information available |
| <b>Bulk density</b>             | No information available |
| <b>Particle characteristics</b> | No information available |

## Section 10: Stability and reactivity

### Reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

### Chemical stability

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|

### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Possibility of hazardous reactions**

**Possibility of hazardous reactions** Avoid contact with metals. This product contains Sodium azide. Sodium azide can react with Copper, Brass, Lead, and solder in piping systems to form explosive compounds and toxic gases.

**Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** Metals.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

## Section 11: Toxicological information

**Acute toxicity**

**Information on likely routes of exposure**

**Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms** No information available.

**Acute toxicity**

.

**Numerical measures of toxicity**

No information available

| Chemical name | Oral LD50                                 | Dermal LD50             | Inhalation LC50                     |
|---------------|-------------------------------------------|-------------------------|-------------------------------------|
| Water         | > 90 mL/kg ( Rat )                        | -                       | -                                   |
| Trade secret  | = 8290 mg/kg ( Rat )                      | > 7940 mg/kg ( Rabbit ) | > 0.83 mg/L ( Rat ) 4 h             |
| Trade secret  | = 22 g/kg ( Rat )                         | > 20 g/kg ( Rabbit )    | -                                   |
| Trade secret  | = 5700 mg/kg ( Rat )<br>= 16 g/kg ( Rat ) | -                       | = 320 mg/m <sup>3</sup> ( Rat ) 4 h |
| Trade secret  | = 27 mg/kg ( Rat )                        | = 20 mg/kg ( Rabbit )   | 0.054 - 0.52 mg/L ( Rat ) 4 h       |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

|                                          |                                                                                                 |
|------------------------------------------|-------------------------------------------------------------------------------------------------|
| Serious eye damage/eye irritation        | No information available.                                                                       |
| Respiratory or skin sensitisation        | No information available.                                                                       |
| Germ cell mutagenicity                   | No information available.                                                                       |
| Carcinogenicity                          | No information available.                                                                       |
| Reproductive toxicity                    | No information available.                                                                       |
| STOT - single exposure                   | No information available.                                                                       |
| STOT - repeated exposure                 | No information available.                                                                       |
| Aspiration hazard                        | No information available.                                                                       |
| Data used to identify the health effects | Refer to Section 16 for Key literature references and sources for data used to compile the SDS. |

## Section 12: Ecological information

### Ecotoxicity

|                          |                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------|
| Aquatic ecotoxicity      | The environmental impact of this product has not been fully investigated.                      |
| Unknown aquatic toxicity | 0.095 % of the mixture consists of component(s) of unknown hazards to the aquatic environment. |

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                 | Crustacea |
|---------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Trade secret  | -                    | LC50: =0.8mg/L (96h, Oncorhynchus mykiss)<br>LC50: =0.7mg/L (96h, Lepomis macrochirus)<br>LC50: =5.46mg/L (96h, Pimephales promelas) | -         |

|                               |                                    |
|-------------------------------|------------------------------------|
| Terrestrial ecotoxicity       | There is no data for this product. |
| Persistence and degradability | No information available.          |

### Bioaccumulative potential

|                 |                                    |
|-----------------|------------------------------------|
| Bioaccumulation | There is no data for this product. |
|-----------------|------------------------------------|

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | -0.698                |

### Mobility in soil

**Mobility** No information available.

**Other adverse effects**

No information available.

## Section 13: Disposal considerations

**Disposal methods**

**Waste from residues/unused products** Not applicable.  
Not Hazardous.  
Dispose of in accordance with local regulations.  
Dispose of waste in accordance with environmental legislation.  
Flush pipes with water frequently if discarding solutions containing Sodium azide into metal piping systems.

**Contaminated packaging** Not applicable.  
Not Hazardous.

## Section 14: Transport information

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information

## Section 15: Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National regulations**

**EPA New Zealand HSNO approval code or group standard** To be determined

**National regulations** There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### **International Inventories**

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDL</b>       | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AIIC</b>          | Contact supplier for inventory compliance status. |

#### **Legend:**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

**ENCS** - Japan Existing and New Chemical Substances

**IECSC** - China Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

### **Section 16: Other information**

**Revision date** 04-Oct-2023

**Revision Note** Reformatted and updated existing information.

#### **Key or legend to abbreviations and acronyms used in the safety data sheet**

##### **Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

|                |                             |             |                                  |
|----------------|-----------------------------|-------------|----------------------------------|
| <b>TWA</b>     | TWA (time-weighted average) | <b>STEL</b> | STEL (Short Term Exposure Limit) |
| <b>Ceiling</b> | Maximum limit value         | <b>*</b>    | Skin designation                 |
| <b>C</b>       | Carcinogen                  |             |                                  |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)

U.S. Environmental Protection Agency ChemView Database

European Food Safety Authority (EFSA)

EPA (Environmental Protection Agency)

Acute Exposure Guideline Level(s) (AELG(s))

U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act

U.S. Environmental Protection Agency High Production Volume Chemicals

Food Research Journal

Hazardous Substance Database

International Uniform Chemical Information Database (IUCLID)

National Institute of Technology and Evaluation (NITE)

Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)

NIOSH (National Institute for Occupational Safety and Health)

National Library of Medicine's ChemID Plus (NLM CIP)

National Library of Medicine's PubMed database (NLM PUBMED)

National Toxicology Program (NTP)

New Zealand's Chemical Classification and Information Database (CCID)

Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications

Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme

Organisation for Economic Co-operation and Development Screening Information Data Set

World Health Organization

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**End of Safety Data Sheet**



# SAFETY DATA SHEET

Revision date 04-Oct-2023

Revision Number 1.2

## Section 1: Identification

### Product identifier

**Product Name** Bio-Plex Standard Diluent  
**Catalogue Number(s)** 9703888, 9704424, 171305042, 171304080M, 10022368, 12005853

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of the supplier of the safety data sheet

|                                                                                                      |                                                                                                                                 |                                                                                             |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Supplier</b><br>Bio-Rad Laboratories Inc.<br>1000 Alfred Nobel Drive<br>Hercules, CA 94547<br>USA | <b>Manufacturer</b><br>Bio-Rad Laboratories, Life Science Group<br>2000 Alfred Nobel Drive<br>Hercules, California 94547<br>USA | <b>Importer</b><br>Bio-Rad Laboratories Pty Ltd<br>189 Bush Road<br>Auckland<br>New Zealand |
| <b>Technical Service</b>                                                                             | +64 9 415 2280 or 0508 805 500<br>sales.nz@bio-rad.com                                                                          |                                                                                             |

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Label elements

#### **Hazard statements**

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Other hazards which do not result in classification

Contains animal source material. (Cattle).

## Section 3: Composition/information on ingredients

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name | CAS No    | Weight-% |
|---------------|-----------|----------|
| Water         | 7732-18-5 | 50 - 100 |
| Trade secret  | -         | 20 - 35  |

| Chemical name | CAS No    | Weight-%     |
|---------------|-----------|--------------|
| Trade secret  | -         | 0.3 - 0.99   |
| HBR 1 Blocker | NO-CAS-18 | 0.1 - 0.299  |
| Trade secret  | -         | 0.01 - 0.099 |
| Trade secret  | -         | 0.01 - 0.099 |
| Trade secret  | -         | 0.01 - 0.099 |
| Trade secret  | -         | 0.01 - 0.099 |

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Non-hazardous ingredients | Proprietary | Balance  |

## Section 4: First-aid measures

### Description of first aid measures

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                              |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| Skin contact | Wash skin with soap and water.                                                                                    |
| Ingestion    | Rinse mouth.                                                                                                      |

### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

### Indication of any immediate medical attention and special treatment needed

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

### Special protective actions for fire-fighters

|                                                                |                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|

## Section 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures



|                                                                    |                                                                                      |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Personal precautions</b>                                        | Ensure adequate ventilation.                                                         |
| <b>For emergency responders</b>                                    | Use personal protection recommended in Section 8.                                    |
| <b><u>Environmental precautions</u></b>                            |                                                                                      |
| <b>Environmental precautions</b>                                   | See Section 12 for additional Ecological Information.                                |
| <b><u>Methods and material for containment and cleaning up</u></b> |                                                                                      |
| <b>Methods for containment</b>                                     | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>                                     | Pick up and transfer to properly labelled containers.                                |
| <b><u>Precautions to prevent secondary hazards</u></b>             |                                                                                      |
| <b>Prevention of secondary hazards</b>                             | Clean contaminated objects and areas thoroughly observing environmental regulations. |

## Section 7: Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

|                               |                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b>     | For long-term storage, keep the battery at tempratures between 10°C-30°C. Store according to product and label instructions. |
| <b>Incompatible materials</b> | Metals.                                                                                                                      |

## Section 8: Exposure controls/personal protection

### Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | New Zealand                                          | Australia                                     | ACGIH TLV                                                                                    | United Kingdom                                                   |
|---------------|------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Trade secret  | Ceiling: 0.11 ppm<br>Ceiling: 0.29 mg/m <sup>3</sup> | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup><br>Sodium azide<br>Ceiling: 0.11 ppm<br>Hydrazoic acid vapor | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.3 mg/m <sup>3</sup><br>Sk* |

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** No special protective equipment required.

**Hand protection** No special protective equipment required.

|                                        |                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | No special protective equipment required.                                                                                                                                |
| <b>Respiratory protection</b>          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                |

## Section 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                                | <u>Values</u>             | <u>Remarks • Method</u> |
|------------------------------------------------|---------------------------|-------------------------|
| <b>pH</b>                                      | 7.4                       |                         |
| <b>Melting point / freezing point</b>          | No data available         | None known              |
| <b>Initial boiling point and boiling range</b> | 100 °C                    |                         |
| <b>Flash point</b>                             | No data available         | None known              |
| <b>Evaporation rate</b>                        | No data available         | None known              |
| <b>Flammability</b>                            | No data available         | None known              |
| <b>Flammability Limit in Air</b>               |                           | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available         |                         |
| <b>Lower flammability or explosive limits</b>  | No data available         |                         |
| <b>Vapour pressure</b>                         | No data available         | None known              |
| <b>Relative vapour density</b>                 | No data available         | None known              |
| <b>Relative density</b>                        | No data available         | None known              |
| <b>Water solubility</b>                        | Miscible in water         |                         |
| <b>Solubility(ies)</b>                         | No data available         | None known              |
| <b>Partition coefficient</b>                   | No data available         | None known              |
| <b>Autoignition temperature</b>                | No data available         | None known              |
| <b>Decomposition temperature</b>               |                           | None known              |
| <b>Kinematic viscosity</b>                     | No data available         | None known              |
| <b>Dynamic viscosity</b>                       | No data available         | None known              |
| <b>Explosive properties</b>                    | No information available. |                         |
| <b>Oxidising properties</b>                    | No information available. |                         |

### Other information

|                                 |                          |
|---------------------------------|--------------------------|
| <b>Softening point</b>          | No information available |
| <b>Molecular weight</b>         | No information available |
| <b>VOC content</b>              | No information available |
| <b>Liquid Density</b>           | No information available |
| <b>Bulk density</b>             | No information available |
| <b>Particle characteristics</b> | No information available |

## Section 10: Stability and reactivity

### Reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

### Chemical stability

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|

### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

**Possibility of hazardous reactions**

**Possibility of hazardous reactions** Avoid contact with metals. This product contains Sodium azide. Sodium azide can react with Copper, Brass, Lead, and solder in piping systems to form explosive compounds and toxic gases.

**Conditions to avoid**

**Conditions to avoid** None known based on information supplied.

**Incompatible materials**

**Incompatible materials** Metals.

**Hazardous decomposition products**

**Hazardous decomposition products** None known based on information supplied.

## Section 11: Toxicological information

**Acute toxicity**

**Information on likely routes of exposure**

**Product Information**

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms** No information available.

**Acute toxicity**

.

**Numerical measures of toxicity**

No information available

| Chemical name | Oral LD50                                 | Dermal LD50             | Inhalation LC50                     |
|---------------|-------------------------------------------|-------------------------|-------------------------------------|
| Water         | > 90 mL/kg ( Rat )                        | -                       | -                                   |
| Trade secret  | = 8290 mg/kg ( Rat )                      | > 7940 mg/kg ( Rabbit ) | > 0.83 mg/L ( Rat ) 4 h             |
| Trade secret  | = 22 g/kg ( Rat )                         | > 20 g/kg ( Rabbit )    | -                                   |
| Trade secret  | = 5700 mg/kg ( Rat )<br>= 16 g/kg ( Rat ) | -                       | = 320 mg/m <sup>3</sup> ( Rat ) 4 h |
| Trade secret  | = 27 mg/kg ( Rat )                        | = 20 mg/kg ( Rabbit )   | 0.054 - 0.52 mg/L ( Rat ) 4 h       |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Skin corrosion/irritation** No information available.

|                                                 |                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Serious eye damage/eye irritation</b>        | No information available.                                                                       |
| <b>Respiratory or skin sensitisation</b>        | No information available.                                                                       |
| <b>Germ cell mutagenicity</b>                   | No information available.                                                                       |
| <b>Carcinogenicity</b>                          | No information available.                                                                       |
| <b>Reproductive toxicity</b>                    | No information available.                                                                       |
| <b>STOT - single exposure</b>                   | No information available.                                                                       |
| <b>STOT - repeated exposure</b>                 | No information available.                                                                       |
| <b>Aspiration hazard</b>                        | No information available.                                                                       |
| <b>Data used to identify the health effects</b> | Refer to Section 16 for Key literature references and sources for data used to compile the SDS. |

## Section 12: Ecological information

### Ecotoxicity

**Aquatic ecotoxicity** The environmental impact of this product has not been fully investigated.

**Unknown aquatic toxicity** 0.018 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                 | Crustacea |
|---------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Trade secret  | -                    | LC50: =0.8mg/L (96h, Oncorhynchus mykiss)<br>LC50: =0.7mg/L (96h, Lepomis macrochirus)<br>LC50: =5.46mg/L (96h, Pimephales promelas) | -         |

**Terrestrial ecotoxicity** There is no data for this product.

**Persistence and degradability** No information available.

### Bioaccumulative potential

**Bioaccumulation** There is no data for this product.

| Chemical name | Partition coefficient |
|---------------|-----------------------|
| Trade secret  | -0.698                |

### Mobility in soil

**Mobility** No information available.

**Other adverse effects**

No information available.

## Section 13: Disposal considerations

**Disposal methods**

**Waste from residues/unused products** Not applicable.  
Not Hazardous.  
Dispose of in accordance with local regulations.  
Dispose of waste in accordance with environmental legislation.  
Flush pipes with water frequently if discarding solutions containing Sodium azide into metal piping systems.

**Contaminated packaging** Not applicable.  
Not Hazardous.

## Section 14: Transport information

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**

No information available

**Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information

## Section 15: Regulatory information

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**National regulations**

**EPA New Zealand HSNO approval code or group standard** To be determined

**National regulations** There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

**International Regulations**

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

The Rotterdam Convention Not applicable

#### International Inventories

|               |                                                   |
|---------------|---------------------------------------------------|
| NZIoC         | Contact supplier for inventory compliance status. |
| TSCA          | Contact supplier for inventory compliance status. |
| DSL/NDL       | Contact supplier for inventory compliance status. |
| EINECS/ELINCS | Contact supplier for inventory compliance status. |
| ENCS          | Contact supplier for inventory compliance status. |
| IECSC         | Contact supplier for inventory compliance status. |
| KECL          | Contact supplier for inventory compliance status. |
| PICCS         | Contact supplier for inventory compliance status. |
| AIC           | Contact supplier for inventory compliance status. |

#### Legend:

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances  
**ENCS** - Japan Existing and New Chemical Substances  
**IECSC** - China Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances

### Section 16: Other information

Revision date 04-Oct-2023

Revision Note Reformatted and updated existing information.

#### Key or legend to abbreviations and acronyms used in the safety data sheet

##### Legend Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|         |                             |      |                                  |
|---------|-----------------------------|------|----------------------------------|
| TWA     | TWA (time-weighted average) | STEL | STEL (Short Term Exposure Limit) |
| Ceiling | Maximum limit value         | *    | Skin designation                 |
| C       | Carcinogen                  |      |                                  |

#### Key literature references and sources for data used to compile the SDS

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AELG(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
 Organisation for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

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End of Safety Data Sheet

# SAFETY DATA SHEET

Revision date 04-Oct-2023

Revision Number 1.2

## Section 1: Identification

### Product identifier

**Product Name** Streptavidin-PE  
**Catalogue Number(s)** 171304501, 9704418, 9703887, 9703897

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of the supplier of the safety data sheet

|                                                                                                      |                                                                                                                                 |                                                                                             |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Supplier</b><br>Bio-Rad Laboratories Inc.<br>1000 Alfred Nobel Drive<br>Hercules, CA 94547<br>USA | <b>Manufacturer</b><br>Bio-Rad Laboratories, Life Science Group<br>2000 Alfred Nobel Drive<br>Hercules, California 94547<br>USA | <b>Importer</b><br>Bio-Rad Laboratories Pty Ltd<br>189 Bush Road<br>Auckland<br>New Zealand |
| <b>Technical Service</b>                                                                             | +64 9 415 2280 or 0508 805 500<br>sales.nz@bio-rad.com                                                                          |                                                                                             |

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Label elements

#### **Hazard statements**

Not a hazardous substance or mixture according to the Globally Harmonised System (GHS)

### Other hazards which do not result in classification

No information available.

## Section 3: Composition/information on ingredients

The product contains no substances which at their given concentration, are considered to be hazardous to health

| Chemical name | CAS No | Weight-% |
|---------------|--------|----------|
|---------------|--------|----------|

| Chemical name | CAS No    | Weight-%     |
|---------------|-----------|--------------|
| Water         | 7732-18-5 | 50 - 100     |
| Trade secret  | -         | 0.3 - 0.99   |
| Trade secret  | -         | 0.1 - 0.299  |
| Trade secret  | -         | 0.1 - 0.299  |
| Trade secret  | -         | 0.01 - 0.099 |
| Avidin        | 9013-20-1 | 0.01 - 0.099 |

| Chemical name             | CAS No      | Weight-% |
|---------------------------|-------------|----------|
| Non-hazardous ingredients | Proprietary | Balance  |

## Section 4: First-aid measures

### Description of first aid measures

|              |                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------|
| Inhalation   | Remove to fresh air.                                                                                              |
| Eye contact  | Rinse thoroughly with plenty of water for at least 15 minutes, lifting lower and upper eyelids. Consult a doctor. |
| Skin contact | Wash skin with soap and water.                                                                                    |
| Ingestion    | Rinse mouth.                                                                                                      |

### Most important symptoms and effects, both acute and delayed

|                     |                           |
|---------------------|---------------------------|
| Symptoms            | No information available. |
| Effects of Exposure | No information available. |

### Indication of any immediate medical attention and special treatment needed

|                 |                        |
|-----------------|------------------------|
| Note to doctors | Treat symptomatically. |
|-----------------|------------------------|

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

|                              |                                                                                                         |
|------------------------------|---------------------------------------------------------------------------------------------------------|
| Suitable Extinguishing Media | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
|------------------------------|---------------------------------------------------------------------------------------------------------|

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| Large Fire | CAUTION: Use of water spray when fighting fire may be inefficient. |
|------------|--------------------------------------------------------------------|

|                                |                                                                   |
|--------------------------------|-------------------------------------------------------------------|
| Unsuitable extinguishing media | Do not scatter spilled material with high pressure water streams. |
|--------------------------------|-------------------------------------------------------------------|

### Specific hazards arising from the chemical

|                                            |                           |
|--------------------------------------------|---------------------------|
| Specific hazards arising from the chemical | No information available. |
|--------------------------------------------|---------------------------|

### Special protective actions for fire-fighters

|                                                                |                                                                                                 |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Special protective equipment and precautions for fire-fighters | Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear. |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------|

## Section 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures



|                                                                    |                                                                                      |
|--------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Personal precautions</b>                                        | Ensure adequate ventilation.                                                         |
| <b>For emergency responders</b>                                    | Use personal protection recommended in Section 8.                                    |
| <b><u>Environmental precautions</u></b>                            |                                                                                      |
| <b>Environmental precautions</b>                                   | See Section 12 for additional Ecological Information.                                |
| <b><u>Methods and material for containment and cleaning up</u></b> |                                                                                      |
| <b>Methods for containment</b>                                     | Prevent further leakage or spillage if safe to do so.                                |
| <b>Methods for cleaning up</b>                                     | Pick up and transfer to properly labelled containers.                                |
| <b><u>Precautions to prevent secondary hazards</u></b>             |                                                                                      |
| <b>Prevention of secondary hazards</b>                             | Clean contaminated objects and areas thoroughly observing environmental regulations. |

## Section 7: Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice.

### Conditions for safe storage, including any incompatibilities

|                               |                                                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Storage Conditions</b>     | For long-term storage, keep the battery at tempratures between 10°C-30°C. Store according to product and label instructions. |
| <b>Incompatible materials</b> | Metals.                                                                                                                      |

## Section 8: Exposure controls/personal protection

### Control parameters

**Exposure Limits** This product, as supplied, does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Chemical name | New Zealand                                          | Australia                                     | ACGIH TLV                                                                                    | United Kingdom                                                   |
|---------------|------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Trade secret  | Ceiling: 0.11 ppm<br>Ceiling: 0.29 mg/m <sup>3</sup> | Peak: 0.11 ppm<br>Peak: 0.3 mg/m <sup>3</sup> | Ceiling: 0.29 mg/m <sup>3</sup><br>Sodium azide<br>Ceiling: 0.11 ppm<br>Hydrazoic acid vapor | TWA: 0.1 mg/m <sup>3</sup><br>STEL: 0.3 mg/m <sup>3</sup><br>Sk* |

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

### Appropriate engineering controls

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

### Individual protection measures, such as personal protective equipment

**Eye/face protection** No special protective equipment required.

**Hand protection** No special protective equipment required.

|                                        |                                                                                                                                                                          |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | No special protective equipment required.                                                                                                                                |
| <b>Respiratory protection</b>          | No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required. |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                |

## Section 9: Physical and chemical properties

### Information on basic physical and chemical properties

|                        |                          |
|------------------------|--------------------------|
| <b>Physical state</b>  | Liquid                   |
| <b>Appearance</b>      | aqueous solution         |
| <b>Colour</b>          | colourless               |
| <b>Odour</b>           | Odourless.               |
| <b>Odour threshold</b> | No information available |

| <u>Property</u>                                | <u>Values</u>             | <u>Remarks • Method</u> |
|------------------------------------------------|---------------------------|-------------------------|
| <b>pH</b>                                      | 7.4                       |                         |
| <b>Melting point / freezing point</b>          | 0 °C                      |                         |
| <b>Initial boiling point and boiling range</b> | 100 °C                    |                         |
| <b>Flash point</b>                             | No data available         | None known              |
| <b>Evaporation rate</b>                        | No data available         | None known              |
| <b>Flammability</b>                            | No data available         | None known              |
| <b>Flammability Limit in Air</b>               |                           | None known              |
| <b>Upper flammability or explosive limits</b>  | No data available         |                         |
| <b>Lower flammability or explosive limits</b>  | No data available         |                         |
| <b>Vapour pressure</b>                         | No data available         | None known              |
| <b>Relative vapour density</b>                 | No data available         | None known              |
| <b>Relative density</b>                        | No data available         | None known              |
| <b>Water solubility</b>                        | Miscible in water         |                         |
| <b>Solubility(ies)</b>                         | No data available         | None known              |
| <b>Partition coefficient</b>                   | No data available         | None known              |
| <b>Autoignition temperature</b>                | No data available         | None known              |
| <b>Decomposition temperature</b>               |                           | None known              |
| <b>Kinematic viscosity</b>                     | No data available         | None known              |
| <b>Dynamic viscosity</b>                       | No data available         | None known              |
| <b>Explosive properties</b>                    | No information available. |                         |
| <b>Oxidising properties</b>                    | No information available. |                         |

### Other information

|                                 |                          |
|---------------------------------|--------------------------|
| <b>Softening point</b>          | No information available |
| <b>Molecular weight</b>         | No information available |
| <b>VOC content</b>              | No information available |
| <b>Liquid Density</b>           | No information available |
| <b>Bulk density</b>             | No information available |
| <b>Particle characteristics</b> | No information available |

## Section 10: Stability and reactivity

### Reactivity

|                   |                           |
|-------------------|---------------------------|
| <b>Reactivity</b> | No information available. |
|-------------------|---------------------------|

### Chemical stability

|                  |                                 |
|------------------|---------------------------------|
| <b>Stability</b> | Stable under normal conditions. |
|------------------|---------------------------------|

### Explosion data

**Sensitivity to mechanical impact** None.

**Sensitivity to static discharge** None.

#### Possibility of hazardous reactions

**Possibility of hazardous reactions** Avoid contact with metals. This product contains Sodium azide. Sodium azide can react with Copper, Brass, Lead, and solder in piping systems to form explosive compounds and toxic gases.

#### Conditions to avoid

**Conditions to avoid** None known based on information supplied.

#### Incompatible materials

**Incompatible materials** Metals.

#### Hazardous decomposition products

**Hazardous decomposition products** None known based on information supplied.

### Section 11: Toxicological information

#### Acute toxicity

#### Information on likely routes of exposure

#### Product Information

**Inhalation** Specific test data for the substance or mixture is not available.

**Eye contact** Specific test data for the substance or mixture is not available.

**Skin contact** Specific test data for the substance or mixture is not available.

**Ingestion** Specific test data for the substance or mixture is not available.

**Symptoms** No information available.

#### Acute toxicity

.

#### **Numerical measures of toxicity**

No information available

| Chemical name | Oral LD50            | Dermal LD50              | Inhalation LC50               |
|---------------|----------------------|--------------------------|-------------------------------|
| Water         | > 90 mL/kg ( Rat )   | -                        | -                             |
| Trade secret  | = 3 g/kg ( Rat )     | > 10000 mg/kg ( Rabbit ) | > 42 mg/L ( Rat ) 1 h         |
| Trade secret  | = 8290 mg/kg ( Rat ) | > 7940 mg/kg ( Rabbit )  | > 0.83 mg/L ( Rat ) 4 h       |
| Trade secret  | = 27 mg/kg ( Rat )   | = 20 mg/kg ( Rabbit )    | 0.054 - 0.52 mg/L ( Rat ) 4 h |

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Skin corrosion/irritation** No information available.

**Serious eye damage/eye irritation** No information available.

|                                                 |                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Respiratory or skin sensitisation</b>        | No information available.                                                                       |
| <b>Germ cell mutagenicity</b>                   | No information available.                                                                       |
| <b>Carcinogenicity</b>                          | No information available.                                                                       |
| <b>Reproductive toxicity</b>                    | No information available.                                                                       |
| <b>STOT - single exposure</b>                   | No information available.                                                                       |
| <b>STOT - repeated exposure</b>                 | No information available.                                                                       |
| <b>Aspiration hazard</b>                        | No information available.                                                                       |
| <b>Data used to identify the health effects</b> | Refer to Section 16 for Key literature references and sources for data used to compile the SDS. |

## Section 12: Ecological information

### Ecotoxicity

**Aquatic ecotoxicity** The environmental impact of this product has not been fully investigated.

**Unknown aquatic toxicity** 0.02 % of the mixture consists of component(s) of unknown hazards to the aquatic environment.

| Chemical name | Algae/aquatic plants | Fish                                                                                                                                                                                                                                                                                                      | Crustacea                                                                            |
|---------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Trade secret  | -                    | LC50: 5560 - 6080mg/L (96h, Lepomis macrochirus)<br>LC50: =12946mg/L (96h, Lepomis macrochirus)<br>LC50: 6020 - 7070mg/L (96h, Pimephales promelas)<br>LC50: =7050mg/L (96h, Pimephales promelas)<br>LC50: 6420 - 6700mg/L (96h, Pimephales promelas)<br>LC50: 4747 - 7824mg/L (96h, Oncorhynchus mykiss) | EC50: =1000mg/L (48h, Daphnia magna)<br>EC50: 340.7 - 469.2mg/L (48h, Daphnia magna) |
| Trade secret  | -                    | LC50: =0.8mg/L (96h, Oncorhynchus mykiss)<br>LC50: =0.7mg/L (96h, Lepomis macrochirus)<br>LC50: =5.46mg/L (96h, Pimephales promelas)                                                                                                                                                                      | -                                                                                    |

**Terrestrial ecotoxicity** There is no data for this product.

| Chemical name | Earthworm                                                                            | Avian | Honeybees |
|---------------|--------------------------------------------------------------------------------------|-------|-----------|
| Trade secret  | Acute Toxicity: LC50 0.1 - 1 mg/cm <sup>2</sup> (Eisenia foetida, 48 h filter paper) | -     | -         |

**Persistence and degradability** No information available.

**Bioaccumulative potential**

**Bioaccumulation** There is no data for this product.

**Mobility in soil**

**Mobility** No information available.

**Other adverse effects**

No information available.

**Section 13: Disposal considerations****Disposal methods**

**Waste from residues/unused products** Not applicable.  
Not Hazardous.  
Dispose of in accordance with local regulations.  
Dispose of waste in accordance with environmental legislation.  
Flush pipes with water frequently if discarding solutions containing Sodium azide into metal piping systems.

**Contaminated packaging** Not applicable.  
Not Hazardous.

**Section 14: Transport information**

**IATA** Not regulated

**IMDG** Not regulated

**Transport in bulk according to Annex II of MARPOL and the IBC Code**  
No information available

**Special precautions for user**

Please refer to the applicable dangerous goods regulations for additional information

**Section 15: Regulatory information****Safety, health and environmental regulations/legislation specific for the substance or mixture****National regulations**

**EPA New Zealand HSNO approval code or group standard** To be determined

**National regulations** There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for

more information

### International Regulations

**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

### International Inventories

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDL</b>       | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AIIC</b>          | Contact supplier for inventory compliance status. |

#### **Legend:**

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances  
**ENCS** - Japan Existing and New Chemical Substances  
**IECSC** - China Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances

## Section 16: Other information

**Revision date** 04-Oct-2023

**Revision Note** Reformatted and updated existing information.

### **Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend** Section 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

|                |                             |             |                                  |
|----------------|-----------------------------|-------------|----------------------------------|
| <b>TWA</b>     | TWA (time-weighted average) | <b>STEL</b> | STEL (Short Term Exposure Limit) |
| <b>Ceiling</b> | Maximum limit value         | *           | Skin designation                 |
| <b>C</b>       | Carcinogen                  |             |                                  |

### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 EPA (Environmental Protection Agency)  
 Acute Exposure Guideline Level(s) (AEGLS)  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
 Organisation for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

**Disclaimer**

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**End of Safety Data Sheet**